

Reachable Hosts

S.No	IP Address	Status
1	192.168.10.1	Active
2	192.168.10.2	Active
3	192.168.10.3	Active
4	192.168.10.20	Active
5	192.168.10.100	Active

Analysis: Reconnaissance Value

A ping sweep is one of the first steps during the reconnaissance phase of a penetration test or cyberattack. It helps identify which systems are online and responding to ICMP requests.

An attacker can use the results to:

- Identify live hosts on the network.
- Reduce scanning time by targeting only active systems.
- Discover routers, servers, printers, and workstations.
- Plan further attacks such as port scanning, vulnerability scanning, or exploitation.

Knowing which devices are active allows attackers to focus their efforts on systems that are available and potentially vulnerable.

Security Measures to Reduce the Effectiveness of Ping Sweeps

1. Block ICMP Echo Requests

Configure firewalls on hosts and network devices to block or limit ICMP Echo Requests (ping). This prevents attackers from easily identifying active systems.

2. Network Segmentation and Firewall Rules

Divide the network into separate VLANs or subnets and restrict traffic between them using firewall rules. This limits host discovery and prevents attackers from scanning the entire network.

Conclusion

The ping sweep successfully identifies active hosts on the **192.168.10.0/24** subnet. This information is valuable during the reconnaissance phase because it reveals which devices are online and worth investigating further. Organizations can reduce the effectiveness of ping sweeps by blocking unnecessary ICMP traffic and implementing network segmentation with proper firewall policies.